

# SIMATS ENGINEERING

## CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

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### COURSE OUTCOME COVERED

***C03:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

*Assessment Tool Weightage: 40%*

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QUESTIONS: 4

Total Marks:40

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### *Code of Conduct:*

*I X. Joshua (192312188) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*

## Annexure A

# CASE STUDY

### Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If patient has Fever AND Cough  $\rightarrow$  Possible Flu
- ii. If patient has Fever AND Rash  $\rightarrow$  Possible Measles
- iii. If patient has Cough AND Breathlessness  $\rightarrow$  Possible Pneumonia
- Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

### Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I.  $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$  [Rule 1]
- II.  $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$  [Rule 2]
- III.  $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$  [Rule 3]
- IV. Facts:  $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set  $\theta$  (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

### Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1)  $P1: \text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2)  $P2: \text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3)  $P3: \neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts:  $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences  $P1, P2, P3$  and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication  $\rightarrow$  move  $\neg$  inward  $\rightarrow$  distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause  $\square$ . Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1)  $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell } [1,2]$
- 2)  $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell } [1,1]$
- 3)  $\text{Glitter}[x,y] \rightarrow \text{Gold is at } [x,y]$
- 4)  $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

- (a) Construct the agent’s belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent’s percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.
- (c) Evaluate whether the path  $[1,1] \rightarrow [1,2] \rightarrow [2,2]$  is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

| Criteria                | Weight age | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                    | Level 1 – Beginning                                  |
|-------------------------|------------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------------|
| Understanding of Case   | 25         | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |
| Application of Concepts | 25         | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts         | Fails to apply relevant concepts                     |
| Analysis                | 20         | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth       | Weak or no analysis; lacks reasoning                 |
| Solution Design         | 20         | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification | Incorrect or no solution provided                    |
| Clarity                 | 10         | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure           | Poor clarity; difficult to understand                |

# SIMATS ENGINEERING

## CO3 - ASSESSMENT TOOL 1 - CASE STUDY : ANSWER KEY

Course Name: Artificial Intelligence | Course Code: CSA1709

### Case Study 1 — Smart Medical Diagnosis System

#### (a) Propositional Logic Representation

Let the following propositional symbols be defined:

F = Patient has Fever C = Patient has Cough R = Patient has Rash  
B = Patient has Breathlessness Flu = Possible Flu  
Mea = Possible Measles Pne = Possible Pneumonia

The knowledge base rules are represented as:

Rule 1:  $F \wedge C \rightarrow Flu$  Rule 2:  $F \wedge R \rightarrow Mea$  Rule 3:  $C \wedge B \rightarrow Pne$

Facts about Patient A:

F = True, C = True, B = True (R is not stated, so R = Unknown/False)

#### (b) Forward Chaining

Forward chaining starts from the known facts and fires any rule whose premises are fully satisfied, repeating until no new facts can be derived.

Step 1 (Initial facts): F = T, C = T, B = T

Step 2: Test Rule 1 ( $F \wedge C \rightarrow Flu$ ).

F = T and C = T, so the condition is satisfied. FIRE Rule 1  $\rightarrow$  Derived fact: Flu = True

Step 3: Test Rule 2 ( $F \wedge R \rightarrow Mea$ ).

F = T but R is unknown (not asserted), condition NOT satisfied. Rule 2 does not fire.

Step 4: Test Rule 3 ( $C \wedge B \rightarrow Pne$ ).

C = T and B = T, condition satisfied. FIRE Rule 3  $\rightarrow$  Derived fact: Pne = True

Step 5: No further rules can fire (KB exhausted).

**Final derived diagnoses for Patient A:** Possible Flu (Flu) and Possible Pneumonia (Pne). Possible Measles is not supported because Rash was never observed.

#### (c) Backward Chaining — Goal: Prove 'Patient A has Pneumonia' (Pne)

Backward chaining starts from the goal and recursively reduces it to sub-goals until each sub-goal is matched against a known fact.



Goal: Pne

Pne is the consequent of Rule 3:  $C \sqcap B \sqcap \text{Pne}$

$\sqcap$  Sub-goal 1: C (is Cough true?)

$\sqcap$  Sub-goal 2: B (is Breathlessness true?)

Check Sub-goal 1: C is directly listed as a Fact = True  $\sqcap$  PROVED Check Sub-goal 2: B is directly listed as a Fact = True  $\sqcap$  PROVED

Since both sub-goals are proved TRUE, Rule 3 fires and the goal Pne is CONFIRMED.

Goal reduction tree:

Pne

|

(Rule 3:  $C \sqcap B \sqcap \text{Pne}$ )

/\

C B

(Fact=T) (Fact=T)

Both leaf nodes are known facts (True)  $\sqcap$  Goal 'Patient A has Pneumonia' is VERIFIED.

## Case Study 2 — Autonomous Traffic Management Agent

### (a) Unification of Rule 1 with the Facts

Rule 1:  $\Box x \text{ Vehicle}(x) \Box \text{EmergencyType}(x) \Box \text{ClearPath}(x)$  Facts:

$\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})$

Step 1: Unify  $\text{Vehicle}(x)$  with  $\text{Vehicle}(\text{Ambulance})$

$\Box 1 = \{ x / \text{Ambulance} \}$

Step 2: Apply  $\Box 1$  to the second literal  $\text{EmergencyType}(x)$

$\text{EmergencyType}(\text{Ambulance})$

This matches the fact  $\text{EmergencyType}(\text{Ambulance})$  exactly, so  $\Box 1$  remains consistent. Step 3: Most General Unifier

(MGU):  $\Box = \{ x / \text{Ambulance} \}$

Step 4: Applying  $\Box$  to Rule 1 gives the ground instance:  $\text{Vehicle}(\text{Ambulance}) \Box$

$\text{EmergencyType}(\text{Ambulance}) \Box \text{ClearPath}(\text{Ambulance})$  Both antecedents are true facts, so

$\text{ClearPath}(\text{Ambulance})$  is derived.

### (b) Resolution Proof that $\text{Proceed}(\text{CarA})$ is True

Step 1 — Convert all rules to Conjunctive Normal Form (CNF) [ $\neg P \Box Q$  form]:

Rule 1 (CNF):  $\neg \text{Vehicle}(x) \Box \neg \text{EmergencyType}(x) \Box \text{ClearPath}(x)$  Rule 2 (CNF):

$\neg \text{ClearPath}(x) \Box \text{GreenSignal}(x)$

Rule 3 (CNF):  $\neg \text{Vehicle}(x) \Box \neg \text{Behind}(x,y) \Box \neg \text{GreenSignal}(y) \Box \text{Proceed}(x)$

Facts (unit clauses):  $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal (for refutation):  $\neg \text{Proceed}(\text{CarA})$  Step 2

— Resolution steps:

R1: Resolve  $\text{Vehicle}(\text{Ambulance})$  with  $[\neg \text{Vehicle}(x) \Box \neg \text{EmergencyType}(x) \Box$

$\text{ClearPath}(x)]$ ,  $\Box = \{x / \text{Ambulance}\}$

$\Box \neg \text{EmergencyType}(\text{Ambulance}) \Box \text{ClearPath}(\text{Ambulance})$

R2: Resolve  $\text{EmergencyType}(\text{Ambulance})$  with result of R1

$\Box \text{ClearPath}(\text{Ambulance})$

R3: Resolve  $\text{ClearPath}(\text{Ambulance})$  with  $[\neg \text{ClearPath}(x) \Box \text{GreenSignal}(x)]$ ,

$\Box = \{x / \text{Ambulance}\}$

$\Box \text{GreenSignal}(\text{Ambulance})$

R4: Resolve  $\text{GreenSignal}(\text{Ambulance})$  with  $[\neg \text{Vehicle}(x) \Box \neg \text{Behind}(x,y) \Box$

$\neg \text{GreenSignal}(y) \Box \text{Proceed}(x)]$ ,

$\Box = \{y / \text{Ambulance}\}$

$\Box \neg \text{Vehicle}(x) \Box \neg \text{Behind}(x, \text{Ambulance}) \Box \text{Proceed}(x)$

R5: Resolve  $\text{Vehicle}(\text{CarA})$  with result of R4,  $\Box = \{x / \text{CarA}\}$

$\Box \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \Box \text{Proceed}(\text{CarA})$

R6: Resolve  $\text{Behind}(\text{CarA}, \text{Ambulance})$  with result of R5

$\Box \text{Proceed}(\text{CarA})$

R7: Resolve  $\text{Proceed}(\text{CarA})$  with negated goal  $\neg \text{Proceed}(\text{CarA})$

$\Box \blacksquare$  (empty clause — contradiction reached)

Since resolution produced the empty clause, the refutation succeeds and **Proceed(CarA)** is proved **TRUE**.

### (c) Sufficiency for Two Simultaneous Emergency Vehicles

The current KB is **not sufficient** for two simultaneous emergency vehicles. Rule 1 and Rule 2 would independently grant ClearPath / GreenSignal to both ambulances if they both satisfy  $\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$ , which could create a logical conflict (two conflicting green signals at an intersection) since the KB has no notion of priority, conflicting directions, or mutual exclusion.

#### Additional rules/facts needed:

1.  $\text{Priority}(x,y)$  — a priority ordering fact/predicate when two emergency vehicles conflict (e.g., based on arrival time, severity, or vehicle type).
2.  $\text{ConflictingPath}(x,y)$  — a predicate stating that two vehicles' routes intersect.
3. A new rule:  
$$\begin{aligned} & \neg x \wedge y: \text{Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{ConflictingPath}(x, y) \wedge \text{Priority}(x,y) \\ & \neg \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y) \end{aligned}$$

This ensures only the higher-priority vehicle receives ClearPath when two emergency vehicles conflict.

4. A temporal/queueing fact so the lower-priority vehicle is granted ClearPath immediately afterward rather than being blocked indefinitely.

Justification: without these additions the KB is logically monotonic and would derive ClearPath for both vehicles simultaneously, which is unsafe in the real world; the new predicates restore consistency by making the conclusion depend on a resolvable priority relation.

## Case Study 3 — Agricultural Expert Reasoning System

### (a) Conversion of P1, P2, P3 and Facts into CNF

Symbols: D = SoilDry, I = IrrigationNeeded, W = CropWheat, A = ApplyDripMethod, R = CropAtRisk

#### P1: $D \rightarrow I$ (IrrigationNeeded (D $\rightarrow$ I))

Eliminate implication:  $\neg D \vee I$

No negation to move inward; already in CNF. CNF:  $\neg D \vee I$

#### P2: $I \rightarrow W \rightarrow A$

Eliminate implication:  $\neg(I \rightarrow W) \vee A$

Move  $\neg$  inward (De Morgan):  $(\neg I \vee \neg W) \vee A$  Distribute / flatten

disjunction:  $\neg I \vee \neg W \vee A$  CNF:  $\neg I \vee \neg W \vee A$

#### P3: $\neg A \rightarrow W \rightarrow R$

Eliminate implication:  $\neg(\neg A \rightarrow W) \vee R$  Move  $\neg$  inward

(De Morgan):  $(A \vee \neg W) \vee R$

Distribute / flatten disjunction:  $A \vee \neg W \vee R$  CNF:  $A \vee \neg W \vee R$

**Facts:** SoilDry = True  $\rightarrow$  clause:  $D$  | CropWheat = True  $\rightarrow$  clause:  $W$

### (b) Resolution Refutation — Prove 'ApplyDripMethod' (A)

Negate the goal and add it to the CNF knowledge base, then derive the empty clause.

Clause list:

1.  $\neg D \vee I$  (from P1)
2.  $\neg I \vee \neg W \vee A$  (from P2)
3.  $A \vee \neg W \vee R$  (from P3, not needed for this proof)
4.  $D$  (fact)
5.  $W$  (fact)
6.  $\neg A$  (negated goal)

Resolution steps:

Step 1: Resolve (1)  $\neg D \vee I$  with (4)  $D$  to obtain  $I$

Step 2: Resolve  $I$  with (2)  $\neg I \vee \neg W \vee A$  to obtain  $\neg W \vee A$

Step 3: Resolve  $\neg W \vee A$  with (5)  $W$  to obtain  $A$

Step 4: Resolve  $A$  with (6)  $\neg A$  to obtain the empty clause  $\blacksquare$

Reaching the empty clause  $\blacksquare$  proves, by refutation, that **ApplyDripMethod is TRUE**.

Resolution proof tree:

```
D  $\neg D \vee I$ 
  \ /
  I  $\neg I \vee \neg W \vee A$ 
    \ /
     $\neg W \vee A$   $W$ 
      \ /
      A  $\neg A$  (negated goal)
        \ /
         $\blacksquare$ 
```

### (c) Effect of Removing the Fact 'SoilDry'

If SoilDry (D) is removed from the KB, the clause set no longer contains D. Step 1 of the resolution proof (resolving  $\neg D \vee I$  with D to obtain I) can no longer be performed, so **IrrigationNeeded (I) is no longer**



**derivable.** Consequently Step 2 (deriving A from  $\neg I \sqcup \neg W \sqcup A$ ) also fails, so **ApplyDripMethod (A) cannot be proved true or false purely by monotonic resolution — its status becomes logically Unknown.**

Whether CropAtRisk (R) still follows now depends on the reasoning pattern applied:

- Open-World / classical FOL reasoning: A is simply 'not proved', not proved false. Since P3 requires  $\neg A$  as a premise and  $\neg A$  is not provable, R (CropAtRisk) CANNOT be derived. The KB is logically silent on R.
- Closed-World Assumption (CWA) — typical of rule-based / Prolog-style expert systems: anything not provable is assumed false. Since A cannot be proved, CWA assumes  $A = \text{False}$ , hence  $\neg A = \text{True}$ . With  $W = \text{True}$ , P3 ( $\neg A \sqcup W \sqcup R$ ) fires and CropAtRisk (R) DOES follow.

**Conclusion:** Under strict monotonic (open-world) logical reasoning, CropAtRisk no longer follows once SoilDry is removed — the KB becomes incomplete for that inference. Under the Closed-World Assumption commonly used in rule-based expert systems, CropAtRisk is still derived by default negation of ApplyDripMethod.

## Case Study 4 — Wumpus World Knowledge Agent

Percepts: Stench[1,2] = True, Breeze[1,1] = True, Glitter[2,2] = True

### (a) Belief State Construction using Modus Ponens

Modus Ponens form: from P and  $P \rightarrow Q$ , infer Q. Applying each rule to the matching percept:

Premise: Stench[1,2] = True and Rule 1:  $\text{Stench}[1,2] \rightarrow \text{WumpusAdjacent}[1,2]$

□ Derived:  $\text{WumpusAdjacent}[1,2] = \text{True}$

□ Candidate Wumpus cells = neighbours of [1,2] = { [1,1], [1,3], [2,2] }

Premise: Breeze[1,1] = True and Rule 2:  $\text{Breeze}[1,1] \rightarrow \text{PitAdjacent}[1,1]$

□ Derived:  $\text{PitAdjacent}[1,1] = \text{True}$

□ Candidate Pit cells = neighbours of [1,1] = { [1,2], [2,1] }

Premise: Glitter[2,2] = True and Rule 3:  $\text{Glitter}[x,y] \rightarrow \text{GoldAt}[x,y]$

□ Derived:  $\text{GoldAt}[2,2] = \text{True}$

**All derived conclusions:**  $\text{WumpusAdjacent}[1,2]$ ;  $\text{PitAdjacent}[1,1]$ ;  $\text{GoldAt}[2,2]$ ; and since the agent's start cell [1,1] is stipulated safe in the standard Wumpus World setup,  $\neg \text{Wumpus}[1,1] \rightarrow \neg \text{Pit}[1,1] \rightarrow \text{Safe}[1,1]$  (Rule 4) also fires, giving  $\text{Safe}[1,1] = \text{True}$ .

### (b) Forward Chaining for Wumpus and Pit Locations

Step 1: Start with percepts as facts:  $\text{Stench}[1,2]=\text{T}$ ,  $\text{Breeze}[1,1]=\text{T}$ ,  $\text{Glitter}[2,2]=\text{T}$ ,  $\text{Safe}[1,1]=\text{T}$  (start cell).

Step 2: Fire Rule 1 on  $\text{Stench}[1,2] \rightarrow \text{Wumpus}$  is in one of { [1,1], [1,3], [2,2] }.

Step 3: Since  $\text{Safe}[1,1] = \text{True}$  (agent occupies [1,1] unharmed),  $\neg \text{Wumpus}[1,1]$  holds.

Eliminate [1,1] from the Wumpus candidate set  $\rightarrow \text{Wumpus} \rightarrow \{ [1,3], [2,2] \}$ .

Step 4: Fire Rule 2 on  $\text{Breeze}[1,1] \rightarrow \text{Pit}$  is in one of { [1,2], [2,1] }.

Step 5: No further percepts are available at [1,3], [2,1] or [2,2] to eliminate more candidates, so forward chaining halts with two disjunctive (undetermined) possibilities remaining for both the Wumpus and the Pit.

**Result:** Wumpus is possibly at [1,3] or [2,2]; Pit is possibly at [1,2] or [2,1].

### (c) Safety Evaluation of Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

Cell [1,2]: It is a candidate Pit location (derived in step (b)) and Rule 4 requires  $\neg \text{Wumpus}[1,2] \rightarrow \neg \text{Pit}[1,2]$  to assert  $\text{Safe}[1,2]$ . Since  $\neg \text{Pit}[1,2]$  cannot be established (Pit[1,2] is still an open possibility),  $\text{Safe}[1,2]$  CANNOT be derived.

Cell [2,2]: It is a candidate Wumpus location (derived in step (b)) and also holds the Gold (Glitter[2,2]). Since  $\neg \text{Wumpus}[2,2]$  cannot be established,  $\text{Safe}[2,2]$  CANNOT be derived either.

**Conclusion:** Using the knowledge derived in parts (a) and (b), the path [1,1]  $\rightarrow$  [1,2]  $\rightarrow$  [2,2] is **NOT proven safe**. Both intermediate/target cells remain live hazard candidates (a possible Pit at [1,2] and a possible Wumpus at [2,2]). By Rule 4,  $\text{Safe}(x,y)$  requires both  $\neg \text{Wumpus}$  and  $\neg \text{Pit}$  to be logically established, and neither condition is satisfied for these cells. A cautious rational agent should first visit [2,1] and/or [1,3] to gather additional percepts (checking for absence of breeze/stench) to shrink the candidate sets before attempting to reach the gold, rather than risk the direct path

